Milestone 2 Documentation

Project Title: ImpactSense – Earthquake Impact Prediction
Name: [k.Goutham Naik]
Milestone: Milestone 2 – Model Training and Evaluation

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1. Introduction

The objective of this milestone is to train machine learning models that can predict the potential impact or risk level of earthquakes using historical seismic data. By analyzing features such as magnitude, depth, latitude, and longitude, the system aims to classify earthquake events into different risk levels.

This milestone focuses on developing baseline and advanced machine learning models and evaluating their performance using appropriate metrics.

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2. Dataset Description

The dataset used for this project was obtained from Kaggle and contains historical earthquake records.

Each record represents an earthquake event with information related to its geographic location and seismic characteristics.

The main features used in this milestone include:

- Magnitude – Strength of the earthquake
- Depth – Distance of the earthquake below the Earth's surface
- Latitude – Geographic location of the earthquake
- Longitude – Geographic location of the earthquake

A new column called **risk_level** was created to categorize earthquake impact levels.

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3. Data Preprocessing

Before training the models, the dataset was cleaned and prepared.

The preprocessing steps included:

- Removing missing values from the dataset
- Removing duplicate records
- Creating a new target variable called **risk_level** based on earthquake magnitude

The risk levels were categorized as follows:

- Low Risk – Magnitude less than 2.5
- Medium Risk – Magnitude between 2.5 and 4.5

- High Risk – Magnitude greater than 4.5

Since machine learning models require numerical inputs, the risk categories were encoded into numeric values using label encoding.

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4. Feature Selection

The following features were selected as input variables for training the models:

- Magnitude
- Depth
- Latitude
- Longitude

The target variable for prediction is **risk_level**, which represents the earthquake impact category.

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5. Train-Test Split

To evaluate the performance of the models, the dataset was divided into two parts:

- **Training Data (80%)** – Used to train the models
- **Testing Data (20%)** – Used to evaluate model performance on unseen data

This helps ensure that the model can generalize well to new earthquake events.

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6. Baseline Model Training

Two baseline models were implemented to establish an initial performance benchmark.

Logistic Regression

Logistic Regression is a simple classification algorithm used to predict the probability of a class label.

Decision Tree

The Decision Tree model uses a rule-based structure to classify data based on feature conditions.

These baseline models provide an initial understanding of how well the dataset can predict earthquake risk levels.

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7. Advanced Model Training

To improve prediction performance, advanced ensemble learning models were implemented.

Random Forest

Random Forest is an ensemble learning algorithm that combines multiple decision trees to improve prediction accuracy and reduce overfitting.

Gradient Boosting

Gradient Boosting is an advanced machine learning technique that sequentially improves model predictions by correcting errors from previous models.

These models are capable of capturing more complex relationships within the data.

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8. Model Evaluation

The trained models were evaluated using classification performance metrics.

The main evaluation metric used was **Accuracy**, which measures the percentage of correctly predicted earthquake risk levels.

Other evaluation techniques such as confusion matrix and classification reports were also used to analyze model performance.

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9. Results

After training and evaluating multiple models, their performances were compared.

The results showed that advanced ensemble models such as **Random Forest and Gradient Boosting provided better prediction accuracy compared to baseline models**.

This indicates that ensemble models are more effective for earthquake impact prediction due to their ability to capture complex patterns in seismic data.

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10. Conclusion

In this milestone, several machine learning models were developed and trained to classify earthquake risk levels using seismic data.

The preprocessing and feature engineering steps helped prepare the dataset for machine learning algorithms. Baseline models provided initial results, while advanced models significantly improved prediction performance.

The best performing models will be further optimized in the next milestone to enhance prediction accuracy and support real-world disaster management applications.